Volumes (6.2)

What is the volume of a sphere with radius 1?

"Cross section" method of finding volumes.

- Pick a direction, say x.
- At each value of x, find the area $A(x)$ of the cross-section at x.
- Evaluate

$$\int_{x\,min}^{x\,max} A(x)\,dx$$

The region bounded by

$x = 1$ $\quad y = 0$ $\quad y = x^2$

is rotated about the x-axis. Find the volume of the resulting solid.

The region bounded by

$x = 0$ $y = 1$ $y = x^2$

is rotated about the y-axis. Find the volume of the resulting solid.

The region bounded by

$x = 0$ $y = 1$ $y = x^2$

is rotated about the line $x = -1$. Find the volume of the resulting solid.

The region bounded by

$x = 0$ $y = 2x$ $y = x^2$

is rotated about the line $y = -1$. Find the volume of the resulting solid.

What is the volume of a cone with radius r and height h?